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Subject → ~~Mobile Computing~~
Machine Learning (CSE473)

Q.1 mitchell's definition of Machine Learning

↳ A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .

↳ $\langle T, P, E \rangle$ Triples

↳ (a) spam filter;

- Task (T): Classifying incoming emails as spam or ham (non-spam).
- Performance (P): The percentage of emails correctly classified (Classification accuracy/precision).
- Experience (E): A database of historical emails labeled as spam or ham.

(b) Handwriting Recognizer

- ↳ Task (T): Recognizing and Transcribing handwritten characters/words from image inputs into digital text.
- ↳ Performance (P): Percentage of handwritten characters/words correctly recognized.
- ↳ Experience (E) - A dataset of handwritten character images paired with their true textual labels.

Q.2

Feature	supervised	unsupervised	Reinforcement
Goal/Target	map inputs to known outputs / labels	Discover hidden patterns, structure or groupings	maximize cumulative reward over time
Feedback mechanism	explicit ground truth labels provided	No Feedback / labels provided	delayed reward or penalties
Learning strategy	Predict outcomes based on historical pairs	Group or transform features based in on historical pairs	Learn optimal policies via trial-and-error interaction
Primary Tasks	classification Regression	clustering Dimensionality reduction	control, strategy optimization
model Evaluation	Error metrics (MSE, Accuracy, Cross-Entropy)	Cluster metrics (Silhouette Score, Inertia)	Cumulative Return total Reward
Two algorithms	Linear Regression Support vector machine (SVM)	K-Means Clustering Principal Component Analysis (PCA)	Q-Learning, Deep Q-Network (DQN)
Real Application	Email spam Detection	Customer market segmentation	Autonomous vehicle Navigation

Q.3 ① Why squared error is Not used in Logistic Regression

↳ In Logistic regression, the hypothesis function is non-linear due to the sigmoid activation.

$$h_{\theta}(x) = \sigma(\theta^T x) = 1 / (1 + e^{-\theta^T x})$$

If the mean squared error (MSE) cost function $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$ is used with this non-linear hypothesis, the resulting error surface is non-convex.

Non-convex loss landscapes contain numerous local minima and flat regions, preventing standard optimization algorithms like gradient descent from guaranteeing convergence to the global minimum.

② Log-Loss (Binary cross-Entropy) Function

↳ To ensure convexity, logistic regression uses the Log-Loss function.

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)}))]$$

where $h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$, $y^{(i)} \in \{0, 1\}$, is the true label, and m is the total number of samples.

Q.4 - Noisy data refers to random errors or variance in measured variables. the standard techniques for handling noisy data include.

- Binning - Smooths Sorted data by partitioning values into bins (equal-Frequency or equal-width) and consulting neighboring values. values in each bin are replaced by the bin mean, median, or bin boundaries.
- Regression - Fits data into a mathematical equation (such as simple or multiple linear regression) noise is smoothed by replacing raw observed values with the predicted values along the regression line.
- outlier Analysis - Detects data points that fall far outside the normal distribution using clustering techniques distance metrics, or statistical tests (Ex- z-scores, IQR) Detected outliers can then be removed, replaced, or analyzed separately.
- Combined Human-machine Inspection - Automated algorithms flag suspicious inconsistent, or ~~anomaly~~ anomalous data points which are then routed to domain experts (humans) to verify, correct, or discard.

Section - B

Q.5 ① Four data preprocessing Tasks

- ↳ Data Cleaning: Fix missing values, noise, and outliers.
- ↳ Data Integration - combine multiple data ~~structures~~ ^{sources} into one.
- ↳ Data Reduction - Reduce data size (e.g. Feature selection)
- ↳ Data Transformation: Scale / normalize data for modeling.

⑪ six missing value strategies

- ↳ Drop the row
- ↳ Fill manually
- ↳ Replace with Attribute mean/median
- ↳ Replace with a constant
- ↳ Replace with Class-specific mean/median
- ↳ Predict using an ML model.

⑫ mean imputation

↳ Data 22, 28, ?, 34, 36

$$\text{mean} = \frac{22 + 28 + 34 + 36}{4} = \frac{120}{4} = 30 \text{ (imputed value = } \$30)$$

↳ why median is preferred: Income has extreme high outliers. the median isn't affected by outliers where the mean, whereas the mean gets skewed.

Q.6

Given $n = (1, 2, 3, 4, 5)$, $y = (3, 4, 6, 7, 9)$, and $\hat{y} = 1.3 + 1.5n$;

(A) Predictions & Residuals

$$\text{↳ } n=1 \Rightarrow \hat{y} = 2.8 \Rightarrow e_1 = 3 - 2.8 = 0.2$$

$$\text{↳ } n=2 \Rightarrow \hat{y} = 4.3 \Rightarrow e_2 = 4 - 4.3 = -0.3$$

$$\text{↳ } n=3 \Rightarrow \hat{y} = 5.8 \Rightarrow e_3 = 6 - 5.8 = 0.2$$

$$\text{↳ } n=4 \Rightarrow \hat{y} = 7.3 \Rightarrow e_4 = 7 - 7.3 = -0.3$$

$$\text{↳ } n=5 \Rightarrow \hat{y} = 8.8 \Rightarrow e_5 = 9 - 8.8 = 0.2$$

$$\sum e_i = 0.2 - 0.3 + 0.2 - 0.3 + 0.2 = 0 \text{ (verified)}$$

(b) SSE, SST, and R^2

$$\hookrightarrow SSE = (0.2)^2 + (-0.3)^2 + (0.2)^2 + (-0.3)^2 + (0.2)^2 = 0.30$$

$$\hookrightarrow SST = (3 - 5.8)^2 + (4 - 5.8)^2 + (6 - 5.8)^2 + (7 - 5.8)^2 + (9 - 5.8)^2$$

$$\hookrightarrow R^2 = 1 - \frac{0.30}{22.80} \approx 0.9868 \quad = 22.80$$

(c) MSE and R^2 , Interpretation

$$\hookrightarrow MSE = \frac{0.30}{5} = 0.06 \text{ (or } \frac{0.30}{3} = 0.10 \text{ using } n-2 \text{ degrees of Freedom)}$$

R^2 interpretation 98.68% of the total variance in y is explained by the linear model.